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EDUSYNC CAMPUS AI MASTER TRAINING & KNOWLEDGE GUIDE

Curated University Intelligence, Career Advisory, Core CS & KTU Rules

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SECTION 1: KTU ACADEMIC REGULATIONS & DEGREE CLASSIFICATIONS

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* Degree Requirements:
  - B.Tech Regular: 160 170 Credits (depending on scheme).
  - B.Tech Honours / Minor: 185 Credits (170 base + 15 additional credits).
  - B.Tech Honours & Minor: 200 Credits (170 base + 15 + 15).
* Passing Standards:
  - Minimum 40% in End Semester Examination (ESE).
  - Minimum 50% in aggregate (CIE + ESE combined).
* Degree Classifications:
  - First Class with Distinction: CGPA >= 8.0 with zero backlogs across all semesters.
  - First Class: CGPA >= 6.5.
  - Second Class: CGPA < 6.5.
* B.Tech Honours Eligibility (Regulation R 16.8):
  - Must register along with 4th semester registration.
  - Must maintain CGPA >= 8.0 at the end of the 8th semester.
  - Must complete 15 additional credits across S4, S5, S6, and S7 without failing any registere

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SECTION 2: CIE, MARKS & OFFICIAL CGPA FORMULAS

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* Continuous Internal Evaluation (CIE - 50 Marks):
  - Series Exam 1: 15 Marks (Tests Modules 1 & 2)
  - Series Exam 2: 15 Marks (Tests Modules 3 & 4)
  - Micro-Assignments & Quizzes: 10 Marks
  - Class Attendance: 10 Marks
* CGPA to Percentage Conversion Formulas:
  - B.Tech 2024 Scheme: Percentage Marks = 10 CGPA
  - B.Tech 2019 Scheme, MCA 2020, MBA 2020: Percentage Marks = (10 CGPA) - 2.5
  - B.Tech 2015 Scheme, B.Arch 2016: Percentage Marks = (10 CGPA) - 3.75
* SGPA Formula:
  SGPA = SUM(Course Credit Grade Point Earned) / SUM(Total Course Credits Registered)
* Grade Point Scale:
  - S Grade: 10.0 (>=90%)
  - A+ Grade: 9.0 (85 89%)
  - A Grade: 8.5 (80 84%)

- B+ Grade: 8.0 (75 79%)
- B Grade: 7.5 (70 74%)
- C+ Grade: 7.0 (65 69%)
- C Grade: 6.5 (60 64%)
- D Grade: 6.0 (55 59%)
- P Grade: 5.5 (50 54%)
- F Grade: 0.0 (<50% aggregate or <40% in ESE)

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SECTION 3: ATTENDANCE, DUTY LEAVES (OD) & MEDICAL CONDONATION

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* Attendance Thresholds:
  - Regular Requirement: Minimum 75% attendance in each registered course to write ESE.
  - Medical Condonation Zone (60% to 74% - Regulation R 6.2):
    Principal can condone shortage down to 60% on genuine medical grounds (hospitalization/acci
    with medical fitness certificate. Maximum allowed in 2 semesters across the degree.
  - Critical Shortage (< 60%): Awarded 'FE' grade (Failed due to Eligibility). Must repeat cour
  - Semester Promotion Floor (R 7.32): Minimum 40% average attendance required to advance.
* Special Attendance Relaxations:
  - Menstrual Leave: 2% relaxation (minimum requirement drops to 73%).
  - PWD Students: 5% relaxation (minimum requirement drops to 70%).
* Duty Leave (On-Duty / OD):
  - Granted for representing college/university in hackathons, sports, NSS, NCC, or cultural fe
  - Maximum 10% attendance relaxation for state/national representation; 5% for college events.
  - Multi-tier digital approval chain: Student Application -> Faculty Advisor -> HOD -> Princip

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SECTION 4: ENGINEERING BRANCH COMPARISON & CAREER ADVISORY

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* Computer Science & Engineering (CSE) vs. Civil Engineering (CE):
  - Computer Science: Focus on algorithms, software development, cloud computing, AI/ML, and cy
  - Civil Engineering: Focus on physical infrastructure, structural design, geotechnical system
* Electronics & Communication (ECE) vs. CSE:
  - ECE provides dual career paths: Core VLSI/Embedded Systems (Intel, Qualcomm, Texas Instrume
* Mechanical Engineering (ME):
  - Core manufacturing, CAD/CAM, robotics, aerospace, electric vehicles (EV), and PSU jobs (ISR

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SECTION 5: CORE COMPUTER SCIENCE FUNDAMENTALS

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* Database Management Systems (ACID Properties):
  - Atomicity: Transactions are all-or-nothing units.
  - Consistency: Transitions preserve all integrity constraints.
  - Isolation: Concurrent transactions do not read uncommitted dirty data.
  - Durability: Committed updates permanently persist across crashes and power failures.
* Operating Systems:
  - 4 Coffman Deadlock Conditions: Mutual Exclusion, Hold & Wait, No Preemption, Circular Wait.
  - Avoidance: Banker's Algorithm using Available, Allocation, and Max Need matrices.
  - Paging: Non-contiguous memory allocation dividing memory into fixed frames and logical page
* Computer Networks (OSI 7 Layers):
  1. Physical (raw bits) -> 2. Data Link (Ethernet frames/MAC) -> 3. Network (IP packets) ->
  4. Transport (TCP/UDP segments) -> 5. Session (dialogue) -> 6. Presentation (SSL/TLS) -> 7. A

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SECTION 6: DATA STRUCTURES & ALGORITHMS (DSA) QUICK REFERENCE

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* Binary Search:
  - Array MUST be sorted. Repeatedly divides search space by half (`mid = low + (high - low)/2`
  - Time: O(log N) | Space: O(1) iterative.
* Sorting Complexities:
  - Merge Sort: Time O(N log N) all cases | Space O(N) | Stable divide-and-conquer.
  - Quick Sort: Time O(N log N) avg, O(N) worst | Space O(log N) | In-place partition.
  - Heap Sort: Time O(N log N) | Space O(1).
* Dynamic Programming (DP):
  - Used for optimization problems with overlapping subproblems and optimal substructure.
  - Approaches: Top-Down (Memoization) or Bottom-Up (Tabulation).
  - Common Problems: 0/1 Knapsack, Longest Common Subsequence (LCS), Coin Change, Matrix Chain.

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SECTION 7: CAMPUS PLACEMENTS, ATS RESUME SCORING & HR INTERVIEWS

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* Present-Past-Future Framework for "Tell Me About Yourself":
  - Present (30s): Name, college (TKMIT), branch (CSE), GPA, and technical focus (e.g. backend
  - Past (40s): 1 2 key project accomplishments, hackathons, or internships with quantifiable i
  - Future (20s): Why this target company role aligns with your long-term career ambition.
* STAR Method for Behavioral Interview Questions:
  - S (Situation): Set the scene and background context.
  - T (Task): Specify your unique challenge or responsibility.
  - A (Action): Detail the technical and leadership actions YOU took.
  - R (Result): Conclude with quantifiable outcomes (e.g. "Improved throughput by 35%").
* ATS Resume Optimization:
  - Standard single-column clean format (Inter, Arial, Calibri fonts).
  - Include quantifiable metrics in bullet points (e.g. "Developed REST API serving 500 req/sec
  - Include verified GitHub repository links, live project demos, and LeetCode/HackerRank profi

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SECTION 8: B.TECH 2024 CURRICULUM, HONOURS, MINORS & MOOC CREDITS

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* Project-Based Learning (PBL - 16 Credits):
  - Collaborative learning phase + Self-directed phase (3 Lecture + 1 Project hr/week).
* Challenge Courses (Credit-by-Examination):
  - Recognition of Prior Learning (RPL): Pass ESE directly without regular CIA. Marks do not af
* MOOC Courses (NPTEL / SWAYAM):
  - Must be 12 to 14 weeks duration from the approved Board of Studies list.
  - Pass/Fail certifications with no marks receive letter grade 'B+'.
  - Up to 8 credits can be transferred for B.Tech Honours / Minor requirements.
* Activity Points (120 Points total):
  - Regular Students: 120 activity points (40 points from each group).
  - Lateral Entry Students: 90 activity points (30 points from each group).
  - Categories: NSS/NCC, Sports, Arts, Technical paper presentation, Hackathons, Professional b

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SECTION 9: STUDENT GRIEVANCE & E-GOVERNANCE WORKFLOWS

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* Categories: Academic, Examination, Fee, Infrastructure, Hostel, Anti-Ragging.
* Escalation Chain: Faculty Advisor -> Head of Department (HOD) -> Principal -> KTU Ombudsman.
* Anonymous Option: Supported with secure token tracking.
* Resolution SLA: College resolves submitted grievances within 3 to 5 business days.

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SECTION 10: AI AGENT CONVERSATION EVOLUTION, PITFALLS & CORRECTION PLAYBOOK

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CASE STUDY 1: BRANCH & COURSE COMPARISON QUERIES

* User Query: "IS COMPUTER SCIENCE BEST OR CIVIL" / "WHICH IS THE BEST COURSE"
* Incorrect Initial Agent Behavior:
  - The agent attempted strict literal RAG grounding on KTU administrative syllabus tables.
  - Flawed Output: "Computer Science is not a Civil Engineering program. It is a program that f
* Root Cause: Over-constraining system prompts with "Answer strictly on verified excerpts only"
* How It Was Corrected:
  - Upgraded RAG generation prompt with Gemini/Claude reasoning instructions.
  - Implemented multi-dimensional comparison tables: Focus, Industry Demand, Starting Packages

CASE STUDY 2: PERSONAL ACADEMIC QUERY ROUTING

* User Query: "MY CGPA" / "I HAVE 83% IS IT GOOD"
* Incorrect Initial Agent Behavior:
  - The agent returned a dry theoretical extract from the KTU Academic Manual explaining what a
* Root Cause: Routing all user messages to static PDF embeddings without inspecting if the quer
* How It Was Corrected:
  - Added query intent classification `isPersonalStudentQuery(...)` in `AiService.java`.
  - Injected live MySQL JPA repositories (`MarkRepository`, `InternalMarkRepository`, `Attendan
  - Fetches the active student's exact marks, calculates live SGPA/CGPA, checks attendance thre

CASE STUDY 3: GENERAL CS, CODING & INTERVIEW PREPARATION

* User Query: "Explain Binary Search", "ACID Properties in DBMS", "How to answer 'Tell me about
* Incorrect Initial Agent Behavior:
  - The agent either defaulted to a 5-bullet fallback menu or force-linked CS concepts to KTU c
* How It Was Corrected:
  - Injected Gemini 1.5 Pro and Claude 3.5 Sonnet system instructions into Ollama API calls.
  - Standardized on structured Markdown, code blocks with time/space complexity analysis (O(log
  - Seeded dedicated technical knowledge modules in `Campus_AI_Comprehensive_Handbook.csv` and

CASE STUDY 4: CONSOLIDATED MULTI-FORMAT RAG ARCHITECTURE

* Previous Bottlenecks:
  - Confusing duplicate directories (`src/main/resources/ktu-docs` vs `./ktu-docs/`).
  - Inflexible file support (only PDF).
  - Hardcoded `Modelfile` requiring manual terminal compilation after every prompt change.
* Modernized Architecture:
  - Consolidated all documents exclusively into the root `./ktu-docs/` directory.
  - Extended `RagService.java` to dynamically parse `.pdf` (Apache PDFBox), `.xlsx`/`.xls` (Apa
  - System prompts and model parameters now inject dynamically via Spring Boot into Ollama's HT

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END OF CAMPUS AI MASTER TRAINING GUIDE

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